

CMSI 4072 Project Status Sheet Spring 2026

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Date: February 25, 2026

Project: PhishGuard

Period Covered: February 18 – February 25, 2026

Accomplishments since last report:

Finalized high-level system architecture including separation of content script, background service worker, detection engine, and popup UI components.

Implemented initial Chrome extension skeleton using Manifest V3, including service worker registration and message passing between content script and background script.

Developed FeatureExtractor module capable of parsing URL length, detecting IP-based URLs, counting subdomains, and identifying presence of HTTPS/SSL.

Implemented rule-based HeuristicAnalyzer with weighted scoring system and classification thresholds (Safe, Suspicious, Phishing).

Successfully tested message passing from content script to background service worker and verified correct risk score calculation on sample phishing URLs.

Designed user interface layout for popup window including dynamic risk score display and triggered rule explanations.

Created UML diagrams (Use Case, Component, Deployment, Class, Sequence, Activity) to support SDD documentation.

Scheduled tasks to be done by next report:

Implement DOM-based feature extraction for detecting suspicious forms, hidden inputs, and mismatched form action URLs (March 3).

Implement warning overlay injection for high-risk classifications and test overlay rendering on live websites (March 3).

Integrate optional backend logging endpoint to store flagged URLs using REST API with JSON payload (March 5).

Implement local caching of recently analyzed domains to reduce redundant analysis and improve performance (March 5).

Perform latency testing to ensure page analysis adds less than 100ms overhead to page load (March 6).

Conduct testing on at least 20 known phishing and 20 legitimate websites to evaluate detection accuracy and tune scoring weights (March 7).

Noteworthy risks, concerns, or problems:

Risk: Chrome Manifest V3 service worker limitations may cause background script suspension during inactive states.

Impact: Could interrupt analysis if state is not properly persisted.

Mitigation Plan: Implement chrome.storage-based state persistence and ensure event-driven architecture for all detection triggers.

Risk: False positives from heuristic-only detection.

Impact: May reduce user trust in extension if legitimate websites are flagged.

Mitigation Plan: Adjust scoring weights based on test dataset results and potentially incorporate lightweight ML classifier to improve accuracy.

Risk: Backend API implementation may increase development scope and introduce deployment complexity.

Impact: Could delay integration phase if server configuration issues arise.

Mitigation Plan: Complete full client-side detection first. Backend logging will remain optional and implemented only if core detection system is stable by March 5.

Risk: Time constraints due to overlapping coursework deadlines.

Impact: Reduced development hours during peak academic workload.

Mitigation Plan: Prioritize core detection functionality and documentation first; defer advanced ML integration if necessary.

If you want, I can also:

- Make this sound slightly more technical (to impress)
- Make it shorter and tighter
- Or generate the next week's report template so you can reuse it quickly

Just tell me.